

Rajesh Kumar Jha

AI & ML and Python Developer / Software Engineer

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Profile Summary

- A vibrant performance-driven professional with nearly 3 years of experience in product development (backend), Python, SQL, Natural Language Processing, Machine Learning, Deep Learning, AI/ML Ops, NLTK, Docker, AWS, GCP, and Azure.
- Effectively worked on complex project tasks and delivered on time and with quality resourceful in requirement gathering, testing, development, and technical support, deep understanding of technology with a focus on delivering business solutions, and skilful in completing the design and implementing code.
- Proven competency in suggesting appropriate as well as technology-based solutions for enhancing the functional efficiency of the organization and achieving business excellence.
- Leadership skills with the capability to mentor top-performing cross-functional teams

Education

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| • B.E. from BNMIT Bangalore, ECE, 2016 - 2020 | CGPA: 7.65 |
| • XII from CBSE, JNV DARBHANGA, 2015 | Percentage: 84 |
| • X from CBSE Board, JNV DARBHANGA, 2013 | CGPA: 9.6 |

Work Experience

Wipro as an AI/ML & Python Developer

July 2021 – Present

- Participating in functional reviews for making sure that the functionalities defined are testable and a suitable test strategy can be identified.
- Managing requirements gathering, and collation of project functional and technical requirements, following up with clients throughout the project lifecycle to ensure full understanding of needs and concerns.
- Developed APIs (FastApi and Flask), queries for aggregation, sorting, and multiple filters in Elasticsearch and integrated the modules using Python.
- Enhancing modules delivered to production by developing and rewriting new code, implementing logger, and handling the exception to enhance the efficiency of the application.
- End-to-end design, development, and deployment of the project into the cloud platforms using a docker container and documenting every process.
- Perform data extraction, text generation, and text analysis on AWS Sagemaker and build an auto-tag classification model, using a pre-trained Hugging face Transformer model.
- Conducted unit testing while software development, and continuous inspection of code quality using SonarQube to detect and resolve the compilation bug whenever required with proper root cause analysis.
- Performing healthcare data analysis, data visualization, dimensionality reduction, and statistical modelling to identify the factors that impacted the person's medical enrolment, build a dashboard using Power Bi / Tableau which helps businesses and is appreciated for the analysis.
- Maintaining program modules including operational support, problem resolution, production support, preventative and corrective maintenance, and enhancements.
- Updating knowledge by studying development tools, programming techniques, and computing equipment, participating in educational opportunities, reading professional publications, and maintaining personal networks.

- Performing exploratory data analysis, data visualization, and data cleaning, building predictive models, developing Rest API's, and deploying them on Google Cloud virtual machine servers.
- Extract, Transform & Load (ETL) the data on Google storage and BigQuery storage from several data sources using Python, SQL, and Google cloud services such as Dataprep, Dataflow, Google Ads, and Google Analytics.
- Performing the extraction of the text data from PDFs, Docs, Invoices, Forms, Images, etc using cloud Document AI and Python open-source library textract, building a real-time vehicle license plate detection and extracting the text of the license plate.
- Create ML/DL/NLP models, then perform hyperparameter tuning and cross validation, and also uses Google Cloud Services like AI Platform, AutoML and VertexAI to enhance the accuracy. Uses Looker-ML to build a dashboard for total sales forecasting, and Dialogflow to build a question and answering chatbot.
- Optimize the accuracy of the image classification model, appreciated by the client for the improvement.
- Upgrading knowledge by studying cloud technologies and certified with Google Cloud Professional Data Engineer.

Skills

Key skills

- Python – OOPS, FastAPI, Flask, Tornado, Django, Pandas, Numpy, Pytest, Pyspark, Docker, GIT/GitHub, IDE (PyCharm, VSCode), RestApi, HTTP
- Machine Learning – Scikit-Learn, XgBoost, OpenCV, Keras, TensorFlow, Pytorch, Feature engineering, AutoML, OCR, pattern recognition, Clustering,
- Natural Language Processing - NLTK, Spacy Bert, LSTM, Sentence Transformer, Cognitive Services (Azure/GCP), Syntactic and Semantic Analysis
- Data Analysis – Time Series, EDA, SQL, Power Bi, Tableau, Plotly, Looker
- Database - ElasticSearch, PostgreSQL, MySQL
- Cloud - Google Cloud Platforms, AWS, Azure

Technical Skills

- API Design, Development and Deployment - Debugging, Unit Testing, Test Driven Development (TDD), Exception Handling, Logger Implementation, Automation, Jira, CI/CD, VCS
- Performance testing - Apache Jmeter, Sonarqube, Postman
- Data Science - Data Cleaning, Data Visualization, Data Extraction, Deep Learning, Predictive Modeling, Statistical Analysis, Hypothesis Testing, Speech Recognition, Neural Networks
- Analyst - Financial, Research, Healthcare, Business, Time Series
- Deep Learning - Neural Networks, Object Detection, Computer Vision

Research Work

- [An appropriate and cost-effective hospital recommender system for a patient of rural area using deep reinforcement learning](#)
Indian Institute of Technology Kharagpur Guide: Dr. Manoj Kumar Tiwari
- [Housing market prediction problem using different machine learning algorithms: A case study](#)
- [Machine learning approaches to real estate market prediction problem: a case study](#)
- Flight Delay and Cancellation forecasting using Novel Hybrid Machine Learning Approaches